VMware Licensing Guide

Senior SAM / ITAM Consultant Workbook - Broadcom-era VMware, Agreements, ELP, Flexera FNMS

Updated version: includes deeper Broadcom agreement analysis, VCF/VVF/SPD interpretation, Flexera FNMS ELP preparation workflow, worked ELP models, audit defense, renewal strategy, and client/interview-ready language.

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Scope	VMware infrastructure licensing: vSphere, vCenter, ESXi, VVF, VCF, vSAN, NSX, Aria/VCF Operations, license keys/files, agreements, ELP, FNMS preparation
Important note	This is a SAM/ITAM workbook, not legal advice. For customer commitments, always validate against the signed Transaction Document, the applicable Broadcom agreement, and the SPD/SaaS Listing that applies to the specific order and version.

How to Use This Workbook

This guide is designed to be used like a consulting playbook. First, it explains VMware licensing in plain English. Then it moves into the technical licensing logic, agreement review, ELP preparation, Flexera FNMS workflow, audit defense, and renewal optimization.

Reader need	Use these chapters
Understand VMware licensing from zero	Chapters 1 to 5
Prepare a client ELP	Chapters 10 to 14
Use Flexera FNMS for VMware	Chapter 13
Review VMware contracts/agreement terms	Chapters 6 and 7
Defend an audit or internal compliance review	Chapters 16 and 17
Prepare for interviews/client discussions	Chapters 18 and 19

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1. Plain-English VMware Licensing Mental Model

VMware lets a company run many virtual servers on fewer physical servers. A physical server runs ESXi. vCenter manages multiple ESXi hosts. vSAN turns local disks inside ESXi hosts into shared storage. NSX adds software-defined networking. VCF and VVF are Broadcom-era bundles that package multiple VMware components together.

For SAM, VMware licensing is usually not about counting how many virtual machines exist. It is about counting the physical infrastructure that is allowed to run the VMware software: hosts, processors, cores, clusters, storage capacity, and the product bundle deployed.

Simple version: VMware compliance answers four questions. 1) Which VMware product or bundle did the customer buy? 2) Which ESXi hosts are using it? 3) How many physical CPU cores must be licensed? 4) Are vSAN and bundle components used within the allowed scope?

Term	Plain-English meaning	SAM meaning
ESXi host	A physical server running VMware hypervisor	Licensable hardware unit for vSphere/VCF/VVF core counts
vCenter	Management console for ESXi hosts and clusters	Evidence source and separate component entitlement
vSAN	Storage created from disks inside ESXi hosts	Capacity/TiB risk in addition to core licenses
NSX	Network virtualization and security layer	Usually relevant under VCF component scope
VVF	vSphere Foundation bundle for enterprise workload platform	Core-based subscription with included vSAN entitlement
VCF	Full private cloud stack	Core-based subscription with more components and larger vSAN entitlement
SPD	Specific Program Documentation	Product-specific legal use-rights source that can override general terms

The single most useful mental model

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VMware compliance =
Product/bundle entitlement + physical core count + vSAN TiB count + component scope + agreement
restrictions + evidence quality
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This is why a VMware ELP cannot be prepared from a purchase report alone. You need the entitlement side and the technical deployment side. The entitlement side comes from the Broadcom portal, contracts, order forms, transaction documents, support records, and legacy VMware records. The deployment side comes from vCenter, ESXi hardware data, vSAN capacity data, NSX, Aria/VCF Operations, HCX, and inventory tools such as Flexera FNMS.

2. Broadcom-Era VMware Licensing Changes

Broadcom/VMware announced the shift away from perpetual licensing toward subscription software and positioned VMware Cloud Foundation and VMware vSphere Foundation as the key offers. This matters because many legacy SAM habits - socket counting, SnS entitlement review, or standalone SKU mapping - are no longer enough for current renewals. [S1]

Old VMware habit	Current Broadcom-era reality	SAM impact
Count CPU sockets for many vSphere licenses	Core-based licensing is central for VCF/VVF and current vSphere subscriptions	Need core-level host inventory
Perpetual license + SnS renewal	Subscription entitlement and support term become central	Renewal expiry creates entitlement and support risk
Many standalone product SKUs	Portfolio consolidated around VVF and VCF with add-ons	Bundle mapping is more important than isolated app recognition
Separate keys for each component	Solution keys and VCF/VVF 9 license files reduce key complexity	License evidence process changes
Audit mainly document-driven	Version 9 adds regular compliance report obligations	Governance calendar required

vSphere standalone availability warning

VMware states that vSphere Standard and vSphere Enterprise Plus are available as standalone editions only up to the 8 Update 3 release, while vSphere 9.1 features are available as part of VMware vSphere Foundation 9.1 and VMware Cloud Foundation 9.1. For SAM, this means version mapping is not cosmetic; it affects whether a standalone entitlement can support the deployed version. [S8]

What changed most for ELP work

- The ELP must calculate physical cores per CPU and apply the minimum 16-core rule per processor.
- The ELP must separate compute entitlement from vSAN capacity entitlement.
- The ELP must prove that bundled components are used only on or for the licensed cores.
- The ELP must capture agreement restrictions: cloud use, hosting, DIPA, portability, support and reporting.
- The ELP must treat license keys/files as evidence, not as the whole entitlement position.

3. Product and Bundle Map

Before doing any calculation, identify the VMware product family. A wrong product classification creates a wrong ELP even if the math is perfect.

Offer/component	What it is	ELP treatment
vSphere / ESXi	Core virtualization layer running workloads on physical hosts	Count physical cores on licensed ESXi hosts
vCenter Server	Centralized management for hosts and clusters	Validate instance entitlement and version rights
vSAN	Software-defined storage across ESXi hosts	Calculate TiB entitlement and actual raw/licensing-relevant capacity
VVF	Enterprise workload platform bundle around vSphere	Core subscription; included vSAN 0.25 TiB per core; components tied to VVF cores
VCF	Full private cloud stack including compute, storage, networking and operations capabilities	Core subscription; included vSAN 1 TiB per core; component use tied to VCF cores
NSX	Software-defined networking/security	Usually bundled in VCF; validate scope and export restrictions
VCF Operations / Automation / Operations for Networks	Operations, automation and monitoring capabilities	Use only to manage licensed VCF/VVF cores unless specific rights allow otherwise
HCX	Migration/mobility component in some deployments	Validate keys, entitlement source and scope
VKS / Supervisor	Kubernetes/container services on vSphere	Included components have specific support/lifecycle notes
Private AI Foundation	AI add-on/component where applicable	Only when the SKU/transaction document includes entitlement

VCF vs VVF in one practical table

Dimension	VVF	VCF
Primary positioning	vSphere-led enterprise workload platform	Full private cloud / full-stack platform
Metric	Per physical core subscription	Per physical core subscription
Minimum	16 cores per processor	16 cores per processor
Included vSAN	0.25 TiB per VVF core	1 TiB per VCF core
Operations entitlement	Aria/VCF Operations-style entitlement for VVF scope	VCF Operations/Automation/Networks for VCF scope
NSX	Not equivalent to full VCF NSX stack	Included component under VCF scope
Cloud use	VVF SPD restricts deployment on Cloud Services	VCF has portability rules for Certified Cloud Services
Best ELP risk	vSAN overage, monitoring non-VVF cores, cloud/hosting restriction	Core deficit, component scope, portability, NSX/export, reporting

4. Core Counting and License Consumption Formulas

For VCF and VVF, core licensing is based on physical CPU cores across ESXi hosts. Broadcom states that customers must license a minimum of 16 physical cores for each physical CPU/processor, even when the CPU has fewer than 16 cores. [S5]

```
Required licensed cores per CPU = MAX(actual physical cores in that CPU, 16)
Required licensed cores per host = SUM(required licensed cores for all CPUs in the host)
Required licensed cores for environment = SUM(required licensed cores across all licensed ESXi hosts)
```

Host example	Physical hardware	Wrong count	Correct VMware count
Small host	2 CPUs x 8 cores	16 actual cores	32 licensed cores because 2 CPUs x 16 minimum
Normal host	2 CPUs x 16 cores	32 actual cores	32 licensed cores
High-core host	2 CPUs x 32 cores	64 actual cores	64 licensed cores
4-socket host	4 CPUs x 12 cores	48 actual cores	64 licensed cores because 4 CPUs x 16 minimum

Why per-socket calculation matters

A common mistake is summing all physical cores and applying a single minimum once. That is wrong. The 16-core minimum applies per processor. A 4-socket server with 12 cores per CPU has 48 actual cores but needs 64 licensed cores, not 48. Broadcom examples for insufficient capacity use the same logic: 2 CPUs with 8 cores each require 32 cores, and 2 CPUs with 24 cores each require 48 cores. [S15]

Consultant notes for source data

- Do not accept CMDB server model data blindly; validate sockets and cores from vCenter/ESXi/hardware inventory.
- Do not exclude BIOS-disabled cores for VCF/VVF if the applicable SPD says deactivated BIOS cores are still licensable.
- Calculate at host level first, then roll up to cluster, vCenter, site and contract pool.
- Keep separate columns for actual physical cores and licensable cores after minimum rule.
- Record whether the host is in production, DR, test, migration, evaluation, or decommissioning status.

5. vSAN Licensing Deep Dive

vSAN licensing is a separate compliance topic. A customer can be compliant on VMware cores and still be non-compliant on vSAN capacity. VCF includes 1 TiB of vSAN capacity per VCF core. VVF includes 0.25 TiB of vSAN capacity per VVF core. The entitlement can be used only across the cores where the relevant vSphere component is deployed. [S2] [S3] [S5]

```
VCF included vSAN TiB = VCF licensed cores x 1.00 TiB
VVF included vSAN TiB = VVF licensed cores x 0.25 TiB
Extra vSAN required = MAX(actual vSAN TiB - included vSAN TiB, 0)
```

Scenario	Core entitlement	Included vSAN	Actual vSAN	Position
VCF private cloud	320 VCF cores	320 TiB	210 TiB	+110 TiB surplus
VVF enterprise cluster	320 VVF cores	80 TiB	210 TiB	-130 TiB deficit before add-ons
No vSAN enabled	320 VVF cores	80 TiB included	0 TiB	No vSAN need, but included entitlement unused

Raw capacity vs usable capacity

For SAM, the key question is licensing-relevant physical capacity, not just datastore free space. Usable capacity shown to administrators after RAID/FTT/overhead is not always the same as licensable raw capacity. Broadcom recommends using its license calculator and related VMware tooling rather than relying only on manual UI interpretation. [S5] [S6]

vSAN risks a senior consultant should catch

Risk	Why it matters	Control
VVF vSAN overage	0.25 TiB/core is small for storage-heavy clusters	Calculate TiB separately and buy add-on capacity if needed
VCF/VVF vSAN pooled incorrectly	Entitlement must be used across relevant licensed cores	Map vSAN clusters to VCF/VVF core scope
UI capacity used as legal count	UI may show usable datastore capacity rather than licensing count	Use official calculator/tooling and store evidence
Stretched cluster ambiguity	Capacity and host scope can span sites	Document both site and cluster topology
Separate vSAN license mixed with bundle vSAN	Technical and entitlement pooling may not behave as expected	Confirm support guidance and key/license file compatibility

6. VMware Agreement Stack and Document Hierarchy

This is the section missing from many VMware licensing guides. In real SAM work, a VMware ELP is not only a math exercise. It is also an agreement interpretation exercise. The license calculation must be aligned to the contract documents that govern the customer order.

Plain-English: The agreement stack tells you which document wins if two documents say different things. For VMware by Broadcom, you must not rely only on old VMware Product Guide habits. Current Broadcom documents introduce Transaction Documents, SPDs/SaaS Listings, Modules, and the Foundation Agreement.

Current Broadcom order of precedence

The Broadcom Foundation Agreement states that conflicts among documents are resolved in this order: Transaction Document first, then Broadcom DPA where applicable, then the applicable SPD or SaaS Listing, then the relevant Module, then the Foundation Agreement. It also states that conflicting customer purchase order terms are deemed null and void. [S4]

Priority	Document	What to extract for VMware SAM
1	Transaction Document / Quote / Order Form / SOW	Product/SKU, quantity, metric, term, start/end dates, support, cloud rights, special negotiated terms, site or account scope
2	Broadcom Data Processing Addendum where applicable	Usually privacy/data processing; not normally core ELP math but relevant for SaaS and support
3	Specific Program Documentation or SaaS Listing	Product-specific use rights: metric, minimums, components, vSAN, cloud/hosting, compliance reporting, support restrictions
4	Relevant Module such as Software/SaaS/Services Module	General license grant, support/update rights, audit/reporting mechanics
5	Foundation Agreement	Core commercial terms, definitions, limitation, payment, dispute, order of precedence

Legacy VMware agreement pack

For older estates, you may still see VMware General Terms, Software Exhibit, Product Guide, EULA, ELA order forms, Enterprise Purchasing Program records, Support and Subscription records, Customer Connect exports, and legacy license keys. These documents remain important for historical entitlements, downgrade/upgrade rights, support end dates, and what a customer was allowed to run before the Broadcom subscription transition. Current renewals, however, must be checked against the new Broadcom transaction documents and SPDs.

Legacy document	What it usually proves	Consultant warning
ELA / Enterprise agreement order form	Large committed purchase, quantities, term, discounting, renewal language	Do not assume old optional extension or support rights survive unchanged after renewal
Product Guide	Feature/use rights for product/version at that time	Use the version/date applicable to the original order
Software Exhibit / EULA	General software license grant and restrictions	May be superseded by newer Broadcom modules on renewal
SnS records	Support/subscription entitlement and upgrade eligibility	Expired support can block upgrades/downgrades or support claims
Customer Connect license keys	Technical activation and historical quantity	Key quantity is evidence, not full legal entitlement
Purchase order / invoice	Commercial purchase evidence	PO terms may not override governing terms unless accepted under the agreement

SPD version control

The VCF and VVF SPDs state that the SPD version published on the date Broadcom accepts the customer transaction document applies to the software in that transaction document. They also state that if the customer installs a release provided as part of support services, the then-current SPD at the date of installation applies to that release. This is critical for customers moving from v8 to v9 or renewing during a period when Broadcom terms changed. [S2] [S3]

Consultant action: always store the exact PDF copy of the SPD applicable to the order date and the exact SPD applicable to the installed release date if the customer installed a later release under support. Do not rely on a live URL only; Broadcom terms can be updated.

7. Agreement Review Checklist

This chapter converts contract reading into a practical SAM extraction workflow. The goal is not to become a lawyer. The goal is to know which terms affect ELP, audit defense, renewal negotiation, and operational risk.

Contract pack request list

Evidence requested	Ask from	Why it matters
Broadcom transaction document / quote / order	Procurement / reseller	Highest-priority document for product, metric, quantity and term
VCF/VVF SPD applicable to order	Broadcom portal / legal docs archive	Product-specific rules and restrictions
Broadcom End User Agreement / modules	Procurement / legal	Audit, support, reporting, general license terms
Legacy VMware ELA/Product Guide/EULA	SAM/procurement archives	Historical perpetual and upgrade/downgrade evidence
Support and subscription records	Broadcom portal / reseller	Upgrade/downgrade and support eligibility
License key export and license file export	VMware admin / Broadcom portal admin	Technical assignment and capacity evidence
Renewal quotes and concession letters	Procurement	Negotiated rights or exceptions
Cloud provider agreements	Cloud team/procurement	Portability and provider-license distinction
Service provider/hosting contracts	Business/legal	DIPA/hosting restrictions can change compliance conclusion

Terms that directly affect ELP

Agreement topic	What to look for	ELP implication
Authorized use limitation	Quantity, metric, named product, term	Defines entitlement ceiling
License metric	Core, TiB, instance, add-on, SaaS meter	Defines calculation logic
Minimums	16 cores per processor; possible order minimums	Prevents undercounting low-core hardware
Component scope	Same physical cores where vSphere in bundle is deployed	Prevents using VCF/VVF components outside licensed scope
vSAN entitlement	0.25 TiB/core or 1 TiB/core plus add-ons	Separate storage calculation
Cloud services	VVF restrictions; VCF certified cloud portability	Cloud deployment may be prohibited or conditional
Hosting/DIPA	Internal applications vs third-party hosted services	Service-provider usage risk
Migration/recovery/testing	Evaluation mode and 15-day testing/recovery language	DR and migration evidence needed
Compliance reporting	180-day reporting, version 9 obligations	Operational governance requirement
Audit	Notice period, audit expenses, true-up at list price	Audit response and financial exposure

Audit clause: practical reading

Broadcom agreement language provides for audit on 30 days prior written notice, remotely or at customer facilities, and requires cooperation. If audit/self-verification reveals unpaid or unlicensed use, customer must order sufficient offerings and support at Broadcom then-current list price within 30 days after written notification. If underpayment is 10 percent or more, the customer may have to reimburse reasonable audit expenses. [S4]

SAM meaning: VMware compliance gaps are not only technical risks. They can create fast commercial exposure because the agreement language can require prompt ordering at then-current list price.

Agreement review output template

Field	Example value	Consultant note
Governing transaction document	Quote Q-12345 / Order O-67890	Attach PDF and capture acceptance date
Agreement hierarchy confirmed	Transaction Document > DPA > SPD > Module > Foundation	Record source and exceptions
Product entitlement	VMware Cloud Foundation	Do not shorten to vSphere if VCF purchased
Metric	Core	Check if add-ons use TiB or other meter
Quantity	960 cores	Separate purchased vs allocated vs consumed
Term	2026-01-01 to 2029-01-01	Use for support and compliance reporting
Applicable SPD	VCF SPD November 2025	Save exact copy
Compliance reporting trigger	Version 9 deployed, report every 180 days	Assign owner and calendar
Cloud/hosting exceptions	None / negotiated exception	Do not infer rights from architecture alone

8. VCF Licensing Deep Dive

VCF is Broadcom/VMware full-stack private cloud software. The VCF SPD says the software is subscription software licensed per core, with a minimum of 16 cores per processor. It also says all components and capabilities included in the software may only be used on, or for, the same physical cores where the vSphere component in the software core license is deployed. [S2]

VCF component entitlement map

Component/right	Entitlement logic	ELP action
vSphere	1 core metric under VCF core license	Count physical ESXi host cores with 16-core CPU minimum
vSAN	1 TiB per VCF core	Calculate actual vSAN TiB and compare to included + add-on TiB
VCF Operations / Automation / Operations for Networks	1 core; manage VCF cores only, with specific MaaS/log exceptions in SPD	Map monitored/managed objects to VCF licensed cores
vCenter Server	1 instance; tied to management of entitled vSphere component licenses	Validate version and managed estate scope
VKS / Supervisor	1 core with lifecycle/support notes	Confirm deployment and support scope
Private AI Foundation	Only if applicable SKU includes it	Check transaction document SKU; do not assume included in all VCF
NSX	Included component under VCF but with export/control restrictions	Check geography/customer/end-user and use cases

VCF portability

The VCF SPD permits license portability only to Certified Cloud Services, and requires the customer to port the entire set of software components to that certified cloud service. It also states that portability rights do not apply to certain license sources such as Value-Added OEM program offerings and certain third-party service-provider or metal-as-a-service program consumption. [S2]

Question	Why it matters
Is the target cloud on the current Certified Cloud Services list?	If not, portability cannot be assumed
Is the customer still the licensee?	Provider-integrated license models may be different

Question	Why it matters
Was the license obtained through OEM/CSP/MaaS route?	Portability may not apply
Are all VCF components ported together?	Partial component portability is a risk
Is there evidence of license validity requested by provider?	Customer may need ongoing representations

VCF restrictions that create audit findings

- Using VCF Operations/Automation to manage non-VCF licensed cores without a specific right.
- Treating included vSAN capacity as available outside VCF-licensed vSphere cores.
- Porting VCF licenses to a cloud service that is not on the certified list.
- Assuming OEM or service-provider consumption has the same portability rights as direct customer subscription.
- Using NSX in restricted countries or restricted end-user contexts without export compliance review.
- Deploying VCF 9 without a working compliance reporting process.

9. VVF Licensing Deep Dive

VVF is VMware vSphere Foundation. The VVF SPD says VVF is subscription software licensed per core with a minimum of 16 cores per processor. It also says components and capabilities included in VVF may only be used on, or for, the same physical cores where the vSphere component in VVF is deployed. [S3]

VVF component entitlement map

Component/right	Entitlement logic	ELP action
vSphere	1 core	Count physical cores on ESXi hosts licensed under VVF
Aria Suite Standard / operations entitlement	1 core; can only monitor VVF cores	Compare monitoring scope to VVF core estate
vSAN	0.25 TiB per VVF core	Calculate included vSAN and add-ons
vCenter Server	1 instance; centralized management of entitled vSphere component licenses	Validate instance, version, and managed estate
vSphere Kubernetes Service	1 core with support lifecycle notes	Confirm feature use and support assumptions
iSCSI support	Allowed only with physical, non-virtualized servers and specified limits	Check vSAN iSCSI target usage
Cloud Services	VVF must not be deployed on Cloud Services	Flag public cloud / third-party infrastructure deployments

VVF hosting/DIPA restriction

The VVF SPD includes restrictions around use with Cloud Services and hosting for third parties. It allows DIPA-style use for a customer delivering its own internal-purpose application, but prohibits other types of hosting or third-party benefit unless expressly allowed. For service-provider, MSP, BPO, or hosted-solution scenarios, this can be the most important term in the agreement review. [S3]

10. Legacy Licensing and Transition Issues

Many customers have a mixed VMware estate: old perpetual vSphere/vCenter/vSAN keys, SnS, VCF/VVF subscriptions, v8 solution keys, and v9 license files. Your ELP must clearly separate historical rights from current subscription rights.

Legacy item	What to validate	Common issue
Perpetual vSphere keys	Quantity, edition, version, support status	Assuming support renewal or v9 rights exist when they do not
SnS / Support and Subscription	Active end date and upgrade rights	Expired support blocks upgrade/downgrade operations

Legacy item	What to validate	Common issue
Old vCenter instance keys	Version compatibility and assignment	vCenter v8 key not compatible with v7 without downgrade/split
Standalone vSAN	Capacity and key type	Cannot assume clean pooling with bundle vSAN entitlement
Legacy VCF perpetual/subscription	Legacy upgrade restrictions	VCF SPD restricts certain legacy perpetual upgrade rights above version 5.x
ELA commitments	Renewal extension rights and negotiated terms	Broadcom-era renewal may change available SKUs and support approach

Upgrade and downgrade key handling

Broadcom provides portal processes for upgrading and downgrading VMware license keys, and also explains solution keys for VCF/VVF. Starting with VCF 5.1.1 and vSphere 8.0U2b, the vSphere key acts as the solution license key for VCF and VVF; component keys can still appear for brownfield customers. [S13] [S14]

11. License Keys, Solution Keys, License Files and Version 9 Reporting

Do not confuse three different things: entitlement, technical activation, and compliance reporting. Entitlement is what the customer bought. License keys/files are how the product is activated or allocated. Compliance reports are how usage is reported for certain version 9 deployments.

Mechanism	Applies to	SAM action
Classic 25-character license keys	Older vSphere/vCenter/vSAN and many v8 deployments	Export assigned keys and capacity; reconcile to portal
Solution license key	VCF/VVF transition model around VCF 5.1.1 and vSphere 8.0U2b+	Confirm if single key enables solution components
VCF/VVF 9 license file	Version 9 licensing model	Track license file, Business Services Console, allocation and reporting
Connected mode	Non-air-gapped VCF/VVF 9 environments	Validate automatic registration/reporting works
Disconnected mode	Air-gapped environments	Calendar manual upload/reporting process
Compliance report	VCF/VVF version 9 deployments	Submit/upload every 180 days as required by SPD

VCF/VVF 9 reporting obligation

The VCF and VVF SPDs state that customers deploying version 9 or above must provide a regularly scheduled verified compliance report. The VCF/VVF SPDs specify reporting 180 days from license registration and every 180 days thereafter; failure to transmit or upload a timely, unaltered report can degrade/block management-plane functionality and suspend support entitlements including update/upgrade access. [S2] [S3] [S4]

Operational control: create a recurring governance task for 180-day reporting. Do not leave this inside the VMware admin team only; SAM, platform operations, procurement and service owner must all know the deadline.

Version 9 licensing operations

VMware explains that VCF 9.0 uses a single license file and VCF Operations to show license allocation and usage. Licensing can operate in connected mode or disconnected mode; the license file can cover VCF cores, vSAN TiBs, Private AI Foundation, VVF cores and VCF Edge cores, while some add-ons continue to use license keys. [S7]

12. Full VMware ELP Methodology

The ELP process compares entitlement to consumption and explains the delta. The senior-consultant version of the process must be evidence-based, repeatable, and defensible.

Step-by-step ELP workflow

Step	Activity	Output
1	Define scope: legal entity, sites, vCenters, clusters, cloud, timeframe, products	Scope statement and assumptions
2	Collect entitlement: contracts, transaction docs, portal exports, licenses, support dates, SPDs	Entitlement register
3	Collect deployment: vCenter, ESXi, CPU/core, vSAN, NSX, Operations, HCX, cloud	Technical inventory pack
4	Clean and normalize data: remove duplicates, retired hosts, stale VMs, wrong clusters	Reconciled host list
5	Classify product scope: legacy vSphere, VVF, VCF, standalone vSAN, add-ons	Product mapping matrix
6	Calculate core consumption per host using 16-core per CPU rule	Core consumption sheet
7	Calculate vSAN capacity requirement separately	vSAN consumption sheet
8	Map bundled components to licensed cores	Component compliance sheet
9	Compare entitlement vs consumption	ELP summary
10	Review agreement restrictions: cloud, hosting, portability, reporting, support	Contract risk log
11	Prepare remediation and optimization options	Action plan
12	Package evidence for audit or renewal	Evidence folder and executive summary

ELP workbook tabs

Tab	Purpose	Key fields
00_Control	Scope, dates, owners, assumptions	Customer, legal entity, vCenters, period, owner, data date
01_Entitlements	All purchases/subscriptions/contracts	Product, SKU, quantity, metric, start/end, site, source doc
02_Agreement_Terms	License rules extracted from docs	Metric, minimums, component scope, cloud/hosting, reporting
03_vCenter_Hosts	Host-level technical data	vCenter, cluster, host, sockets, cores/socket, required cores
04_vSAN	Storage capacity analysis	Cluster, raw TiB, included TiB, add-on TiB, delta
05_Component_Scope	NSX/Operations/HCX/VKS usage	Instance, managed hosts/clusters, mapped entitlement
06_Flexera_Export	FNMS evidence and cross-check	FNMS ID, inventory date, license key, app, device, cluster
07_RVTools_PowerCLI	Independent technical validation	Host/CPU/vSAN exports
08_ELP	Final entitlement vs consumption	Entitlement, consumption, surplus/deficit, risk
09_Remediation	Action plan	Buy, reassign, decommission, consolidate, evidence owner
10_Audit_Pack	Sources and screenshots	Document name, source, date, file path, hash if used

Data quality tests

Test	Failure symptom	Fix
Every ESXi host has socket/core data	Required cores blank or zero	Re-run vCenter/PowerCLI; validate hardware model
Every host assigned to one product scope	Host counted in both VVF and VCF or neither	Agree cluster mapping with VMware architect
vSAN capacity has source and date	Storage delta disputed	Use Broadcom calculator/tooling and vSAN admin validation
License terms match order date/version	Wrong SPD applied	Attach exact SPD version and transaction date
Retired hosts removed with evidence	Old hosts inflate consumption	Collect decommission date and vCenter removal evidence
Cloud deployments classified	Portability/compliance unknown	Check provider, license source and certified list
FNMS inventory current	Stale cluster/VM links	Use vCenter Inventory Issues report and rerun imports

13. Flexera FNMS ELP Preparation Guide

This chapter is intentionally detailed because VMware ELP work in Flexera/FNMS is where many consultants produce shallow outputs. FNMS should be treated as the evidence and reconciliation platform, while the Broadcom-era VCF/VVF ELP may still require an external calculation workbook for new core/vSAN/bundle rules where tool content or configuration does not fully automate the latest Broadcom terms.

Plain-English: Flexera helps you collect and organize VMware evidence. It can show VMware inventory, license keys, server links, contracts and compliance positions. But for Broadcom VCF/VVF, you must still verify the latest legal formula yourself: 16-core per CPU, vSAN TiB entitlement, component scope, reporting and agreement restrictions.

What FNMS can collect for VMware

Flexera documentation states that the VMware Inventory page lists vSphere, vCenter and vSAN application inventory. It also explains that a vCenter server can manage one or more ESXi hosts, each host can run VMs, and vSAN is integrated into the ESXi hypervisor. [S9]

Flexera also documents that the vCenter inventory connector collects cluster and ESX server information and the links between VMs, hosts and clusters; it warns that inaccurate vCenter import can lead to wrong license consumption calculations. [S11]

The FlexNet Manager Suite datasheet says FlexNet Manager for VMware can discover and inventory vSphere and vCenter datacenter server products, capture license key, metric and assigned capacity, provide visibility into license key assignment, detect overused keys, and compare actual server capacity against assigned capacity. [S12]

FNMS role in VMware ELP

ELP area	Use FNMS for	Still validate outside FNMS
Inventory evidence	vCenter, ESXi, clusters, VM links, VMware application inventory	Completeness against RVTools/PowerCLI and platform team list
License key evidence	Assigned keys, metric, capacity where discovered	Broadcom portal entitlement and license-file position
Contracts/purchases	Contract and license records, purchase linkage, SKU normalization	Transaction document priority and SPD version
Compliance calculation	Legacy VMware and supported license models/product use rights	New VCF/VVF 16-core/vSAN/component rules if not fully automated
Exception handling	Exemptions for allowed non-production or special rights where supported	Whether the exception exists in current Broadcom agreement
Reporting	Export evidence, inventory issue reports, license summaries	Audit narrative and legal interpretation

FNMS setup workflow for VMware inventory

Flexera documents a workflow to collect VMware infrastructure inventory from vCenter or standalone ESXi: configure credentials on the inventory beacon, ensure subnet coverage, create a discovery and inventory rule, and select the option to gather VMware infrastructure inventory with advanced recognition of VMware applications. If connecting to vCenter, you do not need to separately connect to ESXi servers managed by that vCenter. [S10]

Step	FNMS navigation/action	Consultant validation
1	Inventory beacon: configure credentials for each vCenter or standalone ESXi	Use a service account approved by VMware admin; confirm it can read host, cluster and licensing data
2	Discovery & Inventory > Subnets: ensure vCenter/ESXi IPs are covered; /32 can be used	Avoid scanning entire data center when known vCenter IPs are available
3	Create dedicated discovery/inventory action	Do not mix with unrelated scans during ELP if you need clean logs
4	Select Gather VMware infrastructure inventory, with advanced recognition of VMware applications	This is key for VMware-specific inventory
5	Create target for VMware infrastructure inventory	Use site/subnet that contains vCenter or standalone ESXi
6	Run/schedule discovery and inventory rule	Capture run date, status and logs
7	Verify Discovery & Inventory > VMware Inventory	Check vSphere, vCenter, vSAN records and host links
8	Run vCenter Inventory Issues Analysis report	Fix stale links, orphaned VMs, movement not captured, incorrect import

FNMS data fields to export for VMware ELP

Data group	Fields to export	Why needed
vCenter/cluster/host	vCenter, datacenter, cluster, host name, host UUID, inventory date	Scope and duplicate control
Hardware	Manufacturer, model, processor type, sockets, cores, core per socket if available	Core calculation and hardware validation
Virtualization links	VM to host, VM to cluster, last inventory dates	Validate cluster mapping and stale movement
VMware applications	Publisher, application, version, edition, installed on, evidence date	Product recognition and version support
License keys	Key, metric, assigned capacity, installed/assigned host if available	Technical activation evidence
Contracts/purchases	PO, invoice, contract, SKU, quantity, metric, start/end, cost, vendor	Entitlement evidence and renewal
FNMS license position	License name, application linked, entitlement, consumption, exemption, compliance status	Baseline tool position
Inventory issues	Orphaned VMs, missing links, stale dates, import errors	Data quality risk before ELP

Recommended FNMS custom fields for Broadcom-era VMware

Custom field	Where	Purpose
VMware Bundle Scope	Inventory device / license	Legacy vSphere, VVF, VCF, standalone vSAN, add-on
Broadcom Site ID	License/contract	Tie portal entitlement to customer site/account
Transaction Document ID	License/contract	Tie entitlement to highest-precedence document
Applicable SPD Version	License/contract	Record VCF/VVF SPD date/version
License File ID	License/contract	Track VCF/VVF v9 license file
Compliance Report Due Date	License/contract/task	Track 180-day reporting obligation
vSAN Included TiB per Core	License	0.25 for VVF, 1.0 for VCF
VMware Core Minimum Rule	License	Flag 16-core per processor rule
Cloud/Hosting Restriction Reviewed	Contract/license	Evidence that architecture was checked
ELP Treatment	License/device	FNMS automated / manual workbook / excluded with evidence

Building the FNMS license records

In FNMS, the exact UI labels can vary by version and configuration, but the senior workflow is consistent: import purchases/contracts, create or validate license records, link VMware applications, set quantities and dates, run reconciliation, then export results for the external ELP workbook. For Broadcom-era bundles, do not force a legacy processor/socket license model if it does not represent the current contract.

License record	Suggested treatment	Notes
Legacy vSphere Standard/Enterprise Plus	Use FNMS VMware license model/content where supported	Validate version and support status
VVF Core subscription	Create dedicated VVF entitlement record and external core calculation if needed	Metric: cores; apply 16-core/CPU in workbook if FNMS cannot automate
VCF Core subscription	Create dedicated VCF entitlement record and external core/component calculation if needed	Also model vSAN and Operations/NSX scope
vSAN add-on TiB	Separate capacity entitlement record	Do not bury vSAN under core record only
VCF/VVF Operations entitlement	Track as component scope record	Consumption should not be standalone broad monitoring of all cores unless allowed
License file / v9 entitlement	Attach as evidence to contract/license record	Track reporting date and license allocation

FNMS to ELP calculation flow

Flow step	Action	Output
1	Export FNMS VMware Inventory and vCenter issue report	Inventory evidence pack
2	Export FNMS inventory devices for ESXi hosts with hardware fields	Host/core source file
3	Export FNMS license keys/application installs	Technical license evidence
4	Export purchases/contracts/license records	Entitlement source file
5	Join host data to vCenter/cluster scope	One row per host
6	Apply product scope mapping: VVF/VCF/legacy	Mapped host list
7	Calculate required cores using MAX(cores per CPU,16) per CPU	Core consumption result
8	Add vSAN raw/TiB data from approved source	Storage position
9	Compare to FNMS license position and Broadcom entitlement	Reconciled ELP
10	Document differences between FNMS automated position and manual Broadcom calculation	Defensible adjustment log

FNMS quality checks before you trust the VMware data

Quality check	Why it matters	Fix
vCenter import freshness	Stale import causes wrong VM/host/cluster mapping	Rerun beacon rule and check logs
Orphaned VMs	Can distort virtualization relationships for other publishers	Use vCenter Inventory Issues Analysis report
Host socket/core completeness	Core ELP cannot be calculated without hardware data	Validate from vCenter/PowerCLI
Standalone ESXi discovery	Not all hosts are managed by vCenter	Add standalone ESXi targets if applicable
Duplicate hosts	Same host discovered via multiple sources can double count	Normalize host name, UUID, serial, vCenter
Masked or missing license keys	Key evidence incomplete	Review service account permissions and vCenter licensing access
Retired devices still active	Inflates consumption	Update asset lifecycle status and collect decommission evidence
Wrong application mapping	A vCenter/vSAN app record alone may not equal VCF/VVF bundle scope	Use manual bundle mapping field

How to present FNMS limitations without sounding weak

Do not say, "Flexera cannot do VMware." That is too broad and sounds unprofessional. Say this instead:

> "Flexera FNMS is being used as the discovery, inventory, contract and evidence platform for VMware. For the current Broadcom VCF/VVF subscription model, we are additionally applying the latest SPD rules in an external ELP workbook because the compliance conclusion depends on current product-specific terms such as the 16-core-per-processor minimum, included vSAN TiB per core, component scope and version 9 reporting obligations."

FNMS deliverables for a VMware ELP

Deliverable	Contents
FNMS evidence extract pack	VMware Inventory, inventory devices, license keys, contracts, purchases, license records, vCenter issue report
FNMS data quality log	Stale imports, orphaned VMs, duplicate hosts, missing cores, missing keys, remediation actions
FNMS vs manual ELP reconciliation	Where tool position aligns; where manual Broadcom SPD adjustment was applied
Contract linkage pack	Each entitlement record linked to transaction document, support term and SPD
Final VMware ELP workbook	Entitlement, consumption, vSAN, component scope, deficits/surplus, remediation
Audit evidence folder	Exports, screenshots, calculations, assumptions, approval from VMware/platform owner

14. Worked ELP Examples

Example A - VVF estate with vSAN overage

Input	Value
Production hosts	6 hosts, each 2 CPUs x 24 cores
DR hosts	4 hosts, each 2 CPUs x 16 cores
Product scope	VVF
VVF entitlement	500 cores
Actual vSAN raw capacity	180 TiB
Additional vSAN add-on entitlement	100 TiB

```
Production cores = 6 x 2 x 24 = 288
DR cores = 4 x 2 x 16 = 128
Total VVF required cores = 416
Core position = 500 - 416 = +84 cores
Included vSAN = 416 x 0.25 = 104 TiB
Additional vSAN required = 180 - 104 = 76 TiB
Add-on position = 100 - 76 = +24 TiB
```

Conclusion: core position is compliant with 84 surplus cores. vSAN is also compliant only because the customer has 100 TiB add-on vSAN entitlement. Without the add-on, the customer would have a 76 TiB vSAN shortfall.

Example B - VCF estate with core deficit but vSAN surplus

Input	Value
Private cloud hosts	5 hosts, each 2 CPUs x 32 cores
Product scope	VCF
VCF entitlement	300 cores
Actual vSAN capacity	180 TiB

```
VCF required cores = 5 x 2 x 32 = 320
Core position = 300 - 320 = -20 cores
Included vSAN = 320 x 1.0 = 320 TiB
vSAN position = 320 - 180 = +140 TiB
```

Conclusion: the customer is non-compliant due to a 20-core VCF deficit. The vSAN surplus does not offset missing VCF core entitlement. Different metrics cannot be netted against each other.

Example C - Mixed VVF and VCF scope

Cluster	Hosts	Hardware	Scope	Required cores
Cluster A - General VM	8	2 CPUs x 16 cores	VVF	8 x 32 = 256
Cluster B - Private cloud	6	2 CPUs x 32 cores	VCF	6 x 64 = 384
Cluster C - DR	4	2 CPUs x 8 cores	VVF	4 x 32 = 128

Do not simply combine all cores and compare against one bucket. Cluster A and C consume VVF; Cluster B consumes VCF. If Operations, NSX, or vSAN are shared across these clusters, you must validate component scope carefully.

15. Renewal and Optimization Strategy

A VMware renewal is now a commercial, architectural and compliance exercise. The SAM consultant should not only report deficits; they should show options that reduce renewal cost and audit exposure.

Optimization lever	Plain-English logic	SAM calculation impact
Retire unused hosts	Stop paying for hardware that does no useful work	Reduces core count and maybe vSAN capacity
Consolidate workloads	Run same workload on fewer, better-utilized hosts	Reduces licensed host footprint

Optimization lever	Plain-English logic	SAM calculation impact
Separate VCF and VVF clusters	Do not license basic workloads as full VCF if not needed	Avoids over-bundling
Review vSAN architecture	Storage-heavy VVF clusters can become expensive	May reduce add-on TiB need
Remove stale NSX/Operations scope	Do not manage non-entitled cores accidentally	Reduces component risk
Hardware refresh planning	Avoid low-core multi-socket waste	Optimizes 16-core minimum exposure
Cloud portability review	Use VCF portability only where allowed	Avoids prohibited cloud deployments
Use FNMS evidence for negotiation	Show actual need rather than accepting vendor opening quote	Supports data-backed renewal

Renewal questions to ask stakeholders

- Which vCenters/clusters are in renewal scope, and which are being retired before renewal?
- Are we buying VVF or VCF because we need the components, or because the quote defaulted to that bundle?
- How much vSAN raw/licensing capacity is actually needed over the renewal term?
- Which clusters need NSX, Operations, Automation, VKS or Private AI capabilities?
- Is any environment on public cloud, hosted infrastructure, OEM hardware bundle, CSP or MaaS?
- Are we moving to version 9, and if yes, who owns compliance report submission every 180 days?
- Can hardware consolidation reduce core count before the order is signed?
- Does the transaction document include negotiated exceptions that differ from standard SPD assumptions?

16. Audit Defense Pack

A defensible VMware position is evidence-based. It should not depend on verbal statements from infrastructure teams alone.

Evidence folder	Required content
01_Agreements	Transaction documents, Broadcom agreement, modules, applicable SPDs, legacy VMware terms
02_Entitlements	Broadcom portal exports, license keys, license files, support dates, invoices, PO/quotes
03_Technical_Inventory	vCenter exports, ESXi host list, hardware sockets/cores, RVTools, PowerCLI
04_vSAN	vSAN capacity reports, calculator outputs, cluster mapping
05_Components	NSX, VCF Operations, Automation, Operations for Networks, HCX, VKS, PAIF where applicable
06_Flexera	FNMS exports, discovery logs, VMware inventory, vCenter inventory issues, license records
07_ELP_Calculation	Workbook, formulas, assumptions, reconciliations, deltas
08_Remediation	Purchase plan, decommission evidence, reassignment, architecture changes
09_Approvals	Sign-off from SAM, procurement, platform owner and architect

Audit response narrative

"We calculated VMware consumption at the physical ESXi host level, applying the 16-core-per-processor minimum and mapping each host to its product scope: legacy vSphere, VVF or VCF. We calculated vSAN TiB separately using approved VMware/Broadcom tooling and compared it against included and add-on entitlement. We validated bundled component usage for NSX, Operations, Automation, HCX and related components against the physical cores licensed for the applicable bundle. We reconciled contractual entitlement from transaction documents and Broadcom portal exports with technical evidence from vCenter, FNMS and PowerCLI."

Common audit weaknesses

Weakness	Why it fails
Only portal keys provided	Keys do not prove complete legal entitlement or actual deployment scope
Only vCenter screenshots provided	Screenshots do not prove contract rights or support dates
No SPD copy saved	Cannot prove which terms were applied
Core calculation done at estate level	May miss per-processor 16-core minimum
No vSAN evidence	Storage entitlement may be the real shortfall
FNMS stale import not remediated	Tool position can be challenged
Cloud/hosting ignored	Architecture could violate use restrictions even if core count is sufficient
No v9 reporting evidence	Operational/support risk remains unresolved

17. Governance Operating Model

VMware licensing cannot be controlled only once a year. Host additions, hardware refresh, vSAN expansion, cloud moves, and version 9 reporting can change compliance quickly.

Role	Responsibility
SAM/ITAM owner	Own ELP, entitlement records, compliance reporting governance, evidence pack
VMware platform owner	Own vCenter/ESXi/vSAN/NSX/Operations deployment data and changes
Infrastructure architect	Approve cluster design, VCF/VVF scope, consolidation and cloud decisions
Procurement/vendor manager	Own quotes, transaction documents, renewals, reseller communications
Legal/contracts	Review agreement hierarchy, hosting/cloud exceptions, audit response
Finance	Validate cost allocation, renewal budget and optimization savings
Security/export	Review NSX/export sensitive use cases where relevant
Service owner	Confirm business criticality and decommission/migration acceptance

Change control triggers

Trigger	SAM action
New ESXi host added	Calculate cores before deployment; confirm product scope
CPU or server hardware refresh	Recalculate 16-core minimum impact
New vSAN disks/capacity	Recalculate TiB and add-on need
NSX/Operations scope expanded	Check component licensed-core scope
Move to VCF/VVF 9	Register license file and 180-day report calendar
Cloud migration / hyperscaler use	Validate VCF portability or VVF restriction
Hosting/new external customer workload	Review DIPA/hosting restrictions
Renewal quote received	Rebuild ELP before accepting SKU/quantity

18. Client-Ready and Interview-Ready Language

Client-ready explanation

"VMware licensing has moved from a relatively simple socket/perpetual model to a subscription and physical-core model. For VCF and VVF, we need to count physical cores on every ESXi host, apply the 16-core minimum per processor, and then separately assess vSAN capacity and bundled component usage. The legal position also depends on the Transaction Document and the applicable Specific Program Documentation, not just the license keys visible in vCenter."

Interview answer: How do you prepare a VMware ELP?

"I start by separating entitlement and consumption. On the entitlement side, I collect Broadcom portal exports, transaction documents, license keys or license files, support dates and the applicable SPD. On the consumption side, I collect vCenter/ESXi inventory, CPU sockets and cores, vSAN capacity, NSX, Operations and other bundle-component usage. For VCF/VVF, I calculate required cores host by host using the 16-core-per-processor minimum, calculate vSAN separately, and map components only to licensed bundle scope. If FNMS is available, I use it for VMware inventory, license-key evidence, contracts and data quality checks, then reconcile FNMS output against the latest Broadcom terms in the ELP workbook."

Interview answer: How do you use Flexera FNMS for VMware?

"I use FNMS as the discovery, inventory and reconciliation backbone. First I configure VMware inventory through the inventory beacon and vCenter credentials, ensuring subnets and discovery rules are correct. Then I verify VMware Inventory, host and cluster relationships, license key data, contracts, purchases and the vCenter Inventory Issues Analysis report. For current Broadcom VCF/VVF subscriptions, I do not blindly rely on a legacy model. I use FNMS evidence and then apply the current SPD calculation: 16-core CPU minimum, vSAN TiB entitlement, component scope, cloud/hosting restrictions and v9 reporting."

19. What Not to Say and Red Flags

Do not say	Say this instead
VMware is licensed by VM count.	For vSphere/VCF/VVF infrastructure, the key metric is usually physical cores/hosts, not VM count.
vSAN is included, so no issue.	vSAN must be calculated separately; VVF and VCF include different TiB per core.
Flexera gives the final answer automatically.	FNMS provides strong evidence and reconciliation, but current Broadcom terms must be validated against the applicable SPD.
We have keys, so we are compliant.	Keys are technical evidence; compliance requires entitlement, deployment and agreement alignment.
Disabled BIOS cores do not count.	For VCF/VVF, the SPD says BIOS-deactivated cores must be licensed.
VCF can be used in any cloud.	VCF portability is conditional and limited to certified cloud services and eligible license sources.
VVF can run on public cloud if technically possible.	VVF SPD restricts use on Cloud Services.
The customer PO terms override everything.	Broadcom agreement order of precedence can deem conflicting PO terms null and void.
We can ignore v9 reporting.	Version 9 compliance reporting can affect management functionality and support entitlement.

Senior consultant red flags

- Many 2-socket, 8-core hosts: high risk of undercounting because of 16-core CPU minimum.
- vSAN enabled but no TiB analysis: hidden storage compliance risk.
- VCF Operations monitoring non-VCF cores: component scope risk.
- VVF deployed in public cloud or hosted third-party service: cloud/hosting restriction risk.
- License file exists but no 180-day compliance-report owner: operational and support risk.
- FNMS vCenter import stale or incomplete: tool consumption may be wrong.
- Legacy perpetual keys assumed to provide v9 rights: transition/upgrade risk.
- Renewal quote based on all discovered hosts, including retired or migration hosts: avoidable cost risk.
- NSX usage in sensitive geography/end-user context: export compliance review required.
- No saved copy of applicable SPD: weak audit defense.

20. Final Checklists

VMware ELP checklist

Check	Done
Scope agreed: legal entities, sites, vCenters, clusters, products, period	
Transaction documents collected and reviewed	
Applicable VCF/VVF/SPD versions saved	
Broadcom portal entitlement export collected	
License keys/license files exported	
FNMS VMware inventory exported and quality checked	
vCenter/PowerCLI/RVTools host inventory collected	
Socket/core data validated per host	
16-core per processor rule applied	
vSAN TiB measured and reconciled	
NSX/Operations/Automation/HCX/VKS mapped to licensed scope	
Cloud/hosting/DIPA restrictions reviewed	
Version 9 compliance reporting owner and due dates assigned	
Surplus/deficit calculated by product and metric	
Remediation plan documented	
Evidence pack prepared and signed off	

Flexera FNMS checklist

Check	Done
VMware inventory beacon configured with vCenter/ESXi credentials	
vCenter/ESXi subnet coverage confirmed	
Dedicated VMware discovery/inventory action created	
Gather VMware infrastructure inventory option selected	
Inventory rule executed and logs reviewed	
Discovery & Inventory > VMware Inventory reviewed	
vCenter Inventory Issues Analysis report reviewed	
Contracts/purchases/license records linked	
VMware application recognition reviewed	
Custom Broadcom-era fields populated where needed	
FNMS automated position reconciled to manual Broadcom ELP	
FNMS data limitations documented transparently	

Agreement review checklist

Check	Done
Order of precedence understood and documented	
Transaction document captured as highest-priority source	
Product/SKU/metric/quantity/term extracted	
SPD version tied to transaction date and installed release	
Audit clause summarized for stakeholder awareness	
Support/update/upgrade language reviewed	
Cloud portability or restriction reviewed	
Hosting/DIPA wording reviewed	
Migration/recovery/testing rights reviewed	
Benchmarking restrictions reviewed where relevant	
Customer-specific negotiated terms captured	

21. Deep Agreement Interpretation Workbook

This section is a practical clause-by-clause workbook for VMware agreement review. The goal is to convert legal text into SAM actions. For every customer, the actual signed/accepted documents may differ, so this chapter gives a review method rather than pretending one standard agreement covers every case.

21.1 Agreement document inventory - what to request and how to name files

Folder	Document type	File naming convention	Why it matters
01_Governing_Agreement	Broadcom End User Agreement / Foundation Agreement / Software Module	YYYY-MM-DD_Broadcom_EUA_or_Module.pdf	Baseline legal terms, audit, reporting, support, order of precedence
02_Transaction_Docs	Quote, order form, SOW, renewal quote, addendum	YYYY-MM-DD_OrderID_Product_Qty_Metric.pdf	Highest priority document for product, quantity, metric and negotiated terms
03_SPDs	VCF/VVF/SPD or SaaS Listing	YYYY-MM-DD_Product_SPD_EffectiveDate.pdf	Product-specific license metric and restrictions
04_Legacy_VMware	VMware Product Guide, EULA, ELA, SnS records	YYYY-MM-DD_VMware_Legacy_DocType.pdf	Historical rights, transition, downgrade/upgrade evidence
05_Portal_Exports	Broadcom entitlements, keys, license files	YYYY-MM-DD_BroadcomPortal_Export.xlsx	Operational entitlement and key evidence
06_Reseller	Reseller quote, pass-through terms, email confirmations	YYYY-MM-DD_Reseller_OrderConfirmation.pdf	Commercial route and support chain
07_Exception_Approvals	Negotiated exceptions, concession letters, support confirmations	YYYY-MM-DD_Exception_Description.pdf	Can change compliance conclusion if valid and higher-precedence

21.2 Clause-to-ELP mapping

Clause area	How to read it	ELP action	Evidence to store
License grant	Who can use the software, for what purpose and during which term	Confirm legal entity and internal use scope	Agreement page, legal entity name, affiliates language
Metric definition	Core, processor, TiB, instance, SaaS meter, add-on unit	Select correct workbook formula and FNMS license treatment	SPD excerpt, transaction line item
Authorized use limitation	Quantity and unit purchased	Create entitlement row by product and metric	Order/quote line item
Minimums	16 cores per CPU; possible order minimums	Apply calculation floor and separate purchase minimum from consumption	SPD/KB/order guidance
Component scope	Components only on/for same physical cores	Map NSX/Operations/Automation/Aria to licensed host scope	Component inventory and cluster mapping
Storage rights	vSAN TiB per core and add-on rights	Build separate vSAN calculation	vSAN report and entitlement rows
Cloud/portability	Certified cloud list and eligible license source	Classify cloud deployments as allowed/prohibited/conditional	Cloud provider contract and portability evidence
Hosting/DIPA	Internal app vs third-party hosted service	Interview business/service owners	Service catalog, external customer contracts
Migration/recovery/testing	Evaluation mode and 15-day language	Do not count temporary recovery wrongly; keep evidence	DR test calendar and migration plan
Audit/self-verification	Notice, cooperation, purchase obligation and audit expenses	Prepare audit response process and evidence pack	Audit clause copy and response SOP
Compliance reporting	Version 9 report cadence and consequences	Create recurring control	Registration date, report submissions, screenshots
Support/update	Updates/upgrades tied to support and quantity	Validate installed version entitlement	Support end date and entitlement history

21.3 Agreement review questions for legal/procurement

- Is the current transaction document under the Broadcom End User Agreement, an older VMware agreement, or a negotiated enterprise framework?

- Does the order include any negotiated exception to standard SPD terms?
- Does the transaction document list VCF, VVF, standalone vSphere, vSAN add-on, or another SKU? Do not infer from the deployed product name alone.
- Which legal entities and affiliates are covered? Are subsidiaries using the software without being covered?
- Are third-party agents allowed to operate the software on customer behalf, and under what responsibility model?
- Is any workload delivered to external customers or partners? If yes, is it DIPA, hosting, managed service, or third-party benefit?
- Is any VMware software deployed in hyperscaler, hosted cloud, OEM, CSP, or metal-as-a-service environments?
- Are there renewal protections, price holds, optional extensions, or termination rights in the signed agreement?
- Does the support term cover the installed product version and upgrade path?
- Who owns version 9 compliance report submission, and where is evidence stored?

21.4 How to write agreement assumptions professionally

Weak wording	Better wording
We assume the customer has rights.	The ELP uses Transaction Document Q-12345 as the entitlement source for 960 VCF cores for the term 2026-2029. No additional negotiated exceptions were provided at the time of analysis.
Old VMware licenses may cover this.	Legacy perpetual licenses were treated separately and not assumed to grant VCF/VVF 9 usage unless supported by current Broadcom entitlement or explicit upgrade documentation.
vSAN is included.	vSAN entitlement was calculated as 1 TiB per VCF core or 0.25 TiB per VVF core, then compared against measured vSAN capacity.
Cloud use looks fine.	Cloud deployments were not assumed compliant until verified against portability/Cloud Services language and the certified provider/license-source conditions.
FNMS says compliant.	FNMS evidence was reconciled with Broadcom VCF/VVF SPD requirements, including any manual adjustments required for current bundle terms.

22. Flexera FNMS - Click-by-Click VMware ELP Operating Procedure

This operating procedure is designed for a SAM analyst working in FNMS/Flexera One ITAM. Menu names can vary slightly by release and implementation, but the control logic remains the same.

22.1 Prerequisites

Prerequisite	Owner	Validation
FNMS/Flexera One ITAM access with license, contract, purchase and inventory permissions	SAM tool admin	Can view licenses, inventory devices, contracts, reports and discovery rules
Inventory beacon deployed and healthy	SAM tool/platform admin	Beacon visible, last upload successful, services running
vCenter service account available	VMware admin	Can read vCenter, ESXi host hardware, clusters and licensing information
Network connectivity from beacon to vCenter/standalone ESXi	Network/security	Ports open, DNS resolution working, certificates accepted if required
Scope list of vCenters and standalone ESXi hosts	VMware platform owner	Authoritative list signed off
Procurement entitlement pack	Procurement/SAM	Orders, contracts, keys/files, support dates
Independent validation source	VMware admin	RVTools/PowerCLI export for cross-check

22.2 Configure VMware inventory collection

Step	Navigation/action	Expected result	Common mistake
1	Inventory Beacon > Password Manager / credentials area	Credential entry for each vCenter or standalone ESXi	Using a normal personal admin ID instead of service account
2	FNMS > Discovery & Inventory > Subnets	vCenter/ESXi IPs covered by site/subnet; /32 if targeting known servers	Subnet not assigned to correct beacon
3	Discovery & Inventory > Discovery and Inventory Rules > Create Action	Dedicated VMware action created	Combining VMware inventory with broad network discovery during audit crunch
4	Action settings	Gather VMware infrastructure inventory with advanced recognition selected	Selecting only generic inventory and missing VMware relationships
5	Create Target	Target points to vCenter/ESXi site/subnet	Scanning too wide or wrong subnet
6	Create Rule	Rule combines VMware action and target	Rule disabled or schedule not set
7	Run rule manually for ELP baseline	Successful task execution status	Not saving logs/evidence
8	Check beacon logs and upload/import status	Data imported into FNMS	Assuming discovery success means inventory import success

22.3 Validate VMware inventory in FNMS

Validation page/report	What to check	Acceptance criteria
Discovery & Inventory > VMware Inventory	vSphere, vCenter and vSAN application records	All scoped vCenters/hosts represented or explained
Inventory > All Inventory / Inventory Devices	ESXi hosts and vCenter servers	No duplicate hosts; lifecycle status correct
Virtual Devices and Clusters	Cluster, pool, VM and host relationships	VMs linked to current hosts/clusters
License key/application details	VMware license key, metric and assigned capacity where collected	Keys not blank/masked unless explained
vCenter Inventory Issues Analysis	Orphaned VMs, stale host links, missed movements, connector issues	No high-severity unresolved issue before ELP sign-off
Task execution status/logs	Discovery/import errors	Errors remediated or documented with impact

22.4 Create or validate VMware license records

Activity	Detailed instruction	Senior reviewer check
Normalize purchases	Import or manually enter Broadcom/VMware purchases with SKU, PO, quantity, metric, term and cost.	Every purchase line has source file and transaction ID.
Create contracts	Create contract records for Broadcom subscription, legacy ELA/SnS and add-ons.	Contract date, vendor, reseller, legal entity and renewal date populated.
Create license records	Separate license records by product/bundle and metric: VCF Core, VVF Core, vSAN Add-on TiB, legacy vSphere, vCenter if separate.	Do not mix VVF and VCF in one generic vSphere license.
Link applications	Link recognized VMware applications where applicable; for bundle licenses, document application relationship and manual bundle logic.	Application mapping is reviewed against deployed versions.
Attach documents	Attach or reference transaction docs, SPDs, portal exports and license files.	Evidence accessible to audit reviewer.
Set custom fields	Populate Broadcom Site ID, SPD version, license file ID, compliance report due date, bundle scope.	Fields not left blank for current subscriptions.
Run reconciliation	Run FNMS compliance calculation after imports.	Results timestamp captured; differences vs manual ELP explained.

22.5 FNMS report/export pack

Export	Filters	Use in ELP
VMware Inventory	Publisher VMware; current scoped vCenters	VMware application inventory and host evidence
All Inventory Devices	Device role ESXi/vCenter or scoped vCenter names	Host hardware and lifecycle status
Virtual Devices and Clusters	All scoped vCenters/clusters	VM-host-cluster relationship validation
vCenter Inventory Issues Analysis	All scoped vCenters	Data quality remediation
All Licenses	Publisher VMware/Broadcom; active licenses	FNMS compliance baseline
Purchases	Vendor VMware/Broadcom/reseller; active and historical	Entitlement quantities and PO evidence
Contracts	Vendor VMware/Broadcom; active and expired	Term, renewal, support and ELA evidence
Installed Applications	Publisher VMware	Version and edition cross-check
Discovery and Inventory logs	VMware inventory rules	Audit evidence of collection process

22.6 FNMS troubleshooting playbook

Problem	Likely cause	Impact	Fix
vCenter missing	Subnet not covered or beacon not assigned	Hosts/clusters absent from ELP	Add /32 subnet or correct site/beacon target
ESXi hosts not linked to vCenter	vCenter connector failed or stale import	VMware and other publisher calculations wrong	Rerun VMware inventory and review logs
License keys masked or missing	Service account lacks licensing visibility	Key evidence incomplete	Adjust vCenter role/permission through VMware admin
Duplicate hosts	Multiple inventory sources with inconsistent naming	Possible double count	Normalize by host UUID/serial/vCenter and retire duplicate
Retired VMs still linked	Stale virtualization data	Can affect other server publisher positions	Use vCenter Inventory Issues report and rerun import
vSAN visible but no capacity detail sufficient for ELP	FNMS app recognition is not same as raw capacity calculation	Storage shortfall may be missed	Use VMware/Broadcom calculator or PowerCLI vSAN report
FNMS compliance differs from manual ELP	Current Broadcom bundle rules not modeled or custom fields missing	Stakeholder confusion	Create reconciliation tab explaining adjustments
Imports failing intermittently	Credential expiry, cert changes, network firewall, API limits	Stale evidence	Establish monitoring and refresh SOP

23. VMware ELP Excel Workbook Design

Even when FNMS is used, a consultant often needs an external workbook to show the Broadcom-specific calculation clearly. This chapter gives the schema and formulas.

23.1 Host consumption table schema

Column	Example	Rule
Data Source	FNMS / vCenter / PowerCLI	Track where each row came from
vCenter	vc-prod-01	Required for scope and duplicates
Cluster	prod-cluster-a	Needed for product mapping
Host Name	esx-prod-001	One row per physical ESXi host
Host UUID/Serial	host-123/ABC123	Use for dedupe
Lifecycle Status	Active/Retired/Migration	Only active/licensable hosts counted unless evidence supports exclusion
Product Scope	VCF/VVF/Legacy	Driven by contract and architecture
CPU Sockets	2	Physical processors
Cores per CPU	24	Physical cores per processor
Actual Physical Cores	48	Sockets x cores per CPU
Licensable Cores	48	Sockets x MAX(cores per CPU,16)
vSAN Enabled	Yes/No	Drives storage analysis
Raw vSAN TiB	42	Use approved source
Evidence File	PowerCLI_2026-05-29.csv	Audit traceability
Reviewed By	VMware owner name	Sign-off

23.2 Excel formulas

```
Actual_Physical_Cores = CPU_Sockets * Cores_Per_CPU
Licensable_Cores = CPU_Sockets * MAX(Cores_Per_CPU,16)
VCF_Included_vSAN_TiB = SUMIF(Product_Scope,"VCF",Licensable_Cores) * 1.0
VVF_Included_vSAN_TiB = SUMIF(Product_Scope,"VVF",Licensable_Cores) * 0.25
Core_Delta = Entitled_Cores - Consumed_Licensable_Cores
vSAN_Delta = Entitled_vSAN_TiB + Included_vSAN_TiB - Actual_vSAN_TiB
Risk = IF(Core_Delta<0,"High",IF(vSAN_Delta<0,"High","Low/Medium based on component scope"))
```

23.3 Entitlement table schema

Column	Example	Purpose
Product Name	VMware Cloud Foundation	Product/bundle identity
Metric	Core	Calculation unit
Quantity Purchased	960	Entitlement count
Start Date	2026-01-01	Term start
End Date	2029-01-01	Term end/support end
Order/Transaction ID	Q-12345	Highest-priority commercial source
SKU/Part Number	VCF-CORE-SUB...	Procurement traceability
Broadcom Site ID	123456	Portal linkage
Applicable SPD	VCF SPD Nov 2025	Legal rule source
License File/Key	File ID/key folder	Technical activation evidence
vSAN Included Per Core	1.0 TiB	Derived entitlement
Reporting Due Date	2026-07-01	Version 9 control
Evidence Path	/Agreements/...	Audit trail

23.4 Adjustment log schema

Adjustment type	Example	Reason	Evidence
Exclude retired host	esx-old-004	Removed before ELP effective date	Decommission ticket and vCenter removal date
Migration license temporary use	esx-mig-002	Within approved migration window	Migration plan and dates
DR testing treatment	DR cluster used 10 days/year	Within 15-day testing/recovery allowance if applicable	DR test calendar and logs
Manual core override	CPU cores corrected from 0 to 24	FNMS hardware inventory incomplete	PowerCLI and VMware admin sign-off
vSAN manual capacity source	Cluster A capacity set to 120 TiB	FNMS did not provide licensing-relevant raw capacity	VMware calculator output
Bundle scope correction	Cluster B changed VVF to VCF	Architecture owner confirmed VCF workload domain	Architecture diagram and approval

24. Data Collection Scripts and Technical Evidence

This section is not a full automation manual. It gives practical snippets and the evidence logic behind them. Always validate scripts in a non-production session and with VMware administrators.

24.1 PowerCLI host/core export logic

```
Connect-VIServer -Server "vcenter.company.com"

Get-VMHost | ForEach-Object {
$view = Get-View $_.Id
$sockets = $view.Hardware.CpuPkg.Count
$totalCores = $view.Hardware.CpuInfo.NumCpuCores
$coresPerSocket = [math]::Round($totalCores / $sockets,2)
$licensedCores = $sockets * [math]::Max($coresPerSocket,16)
[PSCustomObject]@{
vCenter = $global:DefaultVIServer.Name
Cluster = $_.Parent.Name
Host = $_.Name
Sockets = $sockets
TotalPhysicalCores = $totalCores
CoresPerSocket = $coresPerSocket
RequiredLicensedCores = $licensedCores
Version = $_.Version
Build = $_.Build
}
} | Export-Csv ".\vmware_host_core_export.csv" -NoTypeInformation
```

24.2 Evidence reconciliation checks

Check	Compare	Decision rule
Host count	FNMS vs RVTools vs PowerCLI vs vCenter UI	Differences must be explained before ELP finalization
Core count	FNMS hardware vs PowerCLI vs server model	Use most authoritative recent source and record reason
Cluster mapping	FNMS virtual relationships vs architecture diagram	Architecture owner signs off
License keys	vCenter assigned keys vs Broadcom portal	Missing keys are evidence gap; entitlement still from contract
vSAN capacity	vSAN report/calculator vs UI summary	Use licensing-relevant source
Retired assets	CMDB vs vCenter vs FNMS lifecycle	Exclude only with decommission evidence
Cloud hosts	Cloud provider inventory vs contract	Validate portability/restrictions before inclusion/exclusion

24.3 Data owner sign-off matrix

Data set	Primary owner	Sign-off question
vCenter/ESXi host list	VMware platform owner	Is this the complete active VMware host list for the scope date?
CPU/socket/core values	VMware/admin hardware team	Are these physical CPU details accurate and current?
vSAN capacity	Storage/VMware owner	Is this licensing-relevant capacity, not merely usable/free capacity?
NSX/Operations/HCX	VMware architect	Are these components mapped to the right VCF/VVF scope?
Contracts and entitlements	Procurement/SAM	Are all purchases and renewals included?
Legal terms	Legal/procurement	Are there special terms or exceptions not visible in the order?
FNMS exports	SAM tool owner	Is the inventory current and known issues documented?

25. Scenario Playbook

Scenario 1 - Customer says: "We have enough keys in vCenter"

Response: vCenter keys show technical assignment, not the full legal position. Request the transaction document, Broadcom entitlement export, support term, applicable SPD and host/core inventory. Then reconcile assigned key capacity to legally purchased capacity and deployed host requirements.

Risk	Test	Evidence
Key capacity may not match contract	Compare key folder/export to transaction quantity	Broadcom portal and contract
Wrong version key	Check vCenter/ESXi version vs key version	Portal upgrade/downgrade history
Host requires more cores than key capacity	Apply 16-core/CPU rule	Host core calculation
Standalone key used for bundle component	Check VCF/VVF solution key/component rights	KB/SPD and assigned keys

Scenario 2 - Customer wants to reduce Broadcom renewal quote

Response: Build a pre-renewal ELP and optimization case. Do not negotiate from Broadcom quote alone. Show active hosts, required cores, retirement candidates, vSAN need, VCF vs VVF scope, and component usage. Identify clusters that do not need VCF components and hosts that can be decommissioned before renewal.

Optimization question	Evidence needed	Potential result
Can old hosts be removed before renewal?	Utilization, VM placement, decommission plan	Lower core count
Does every cluster need VCF?	NSX, Automation, Operations, private-cloud requirements	Move some scope to VVF if contractually and technically appropriate
Is vSAN oversized?	Raw capacity, policy, FTT/RAID, growth forecast	Reduce add-on TiB
Are DR hosts always active?	DR design and testing frequency	Correct treatment and evidence
Can hardware refresh reduce socket minimum waste?	Server models and workload capacity	Lower 16-core floor exposure

Scenario 3 - FNMS says compliant but manual ELP shows deficit

Response: Investigate whether FNMS is modeling the same product, metric and current Broadcom terms. FNMS may have accurate inventory but the license record may be legacy, generic, or not configured for current VCF/VVF bundle rules. Prepare a reconciliation note rather than hiding the difference.

Root cause	Fix
License record uses old processor/socket metric	Create/update Broadcom-era core subscription treatment
vSAN not calculated as separate TiB entitlement	Add vSAN tab and entitlement record
FNMS inventory missing recent hosts	Rerun VMware inventory and fix vCenter connector
Manual workbook double-counted retired hosts	Validate lifecycle and exclusion evidence
Product scope mapping wrong	Review cluster architecture with VMware owner

Scenario 4 - VCF/VVF 9 reporting missed

Response: Treat it as an operational compliance incident. Confirm whether the environment is connected or disconnected, obtain the last registration/report date, collect screenshots/logs, submit/upload the required report as per product documentation, and document corrective action to prevent recurrence.

Immediate action	Owner
Check VCF Operations licensing dashboard and registration status	VMware platform owner
Identify last compliance report submission/upload date	SAM + VMware owner
Confirm whether management plane functionality/support access is affected	Platform + support team
Submit/upload report through approved process	VMware owner
Create 180-day recurring control with escalation	SAM governance

Immediate action	Owner
Store evidence and incident closure note	SAM/ITAM owner

26. Workshop Agenda and Stakeholder Questions

26.1 VMware ELP kickoff workshop agenda

Time	Topic	Participants	Output
0-10 min	Scope confirmation	SAM, procurement, VMware owner	Legal entities, vCenters, dates, products
10-25 min	Current architecture overview	VMware architect, platform team	Cluster and product scope map
25-40 min	Entitlement and agreement review	Procurement, legal, SAM	Document request list and ownership
40-55 min	Inventory/FNMS data approach	SAM tool owner, VMware admin	Collection plan and timeline
55-70 min	vSAN and component scope deep dive	Storage, NSX, Operations owners	Data source owners
70-80 min	Cloud/hosting/DR/migration review	Architecture, service owners, legal	Risk items and evidence needs
80-90 min	Next steps and sign-off process	All	RACI, due dates, evidence folder

26.2 Questions for VMware platform team

- How many vCenters are in scope, and are there standalone ESXi hosts not managed by vCenter?
- Which clusters are intended to be VCF, VVF, standalone vSphere, DR, lab, migration or decommissioning?
- Are any hosts temporarily present for migration or hardware refresh?
- Which clusters have vSAN enabled, and what source should be used for licensing-relevant capacity?
- Is NSX deployed? If yes, which clusters/hosts use it and why?
- Which Operations/Aria/VCF Operations tools monitor or manage which hosts/clusters?
- Is HCX used for migration or ongoing mobility?
- Are Supervisor/VKS or Tanzu/Kubernetes features enabled?
- Are any workloads in hyperscaler/cloud/provider-hosted environments?
- Are any workloads delivered to external customers as hosted services?
- What is the planned upgrade path to vSphere/VCF/VVF 9?
- Who owns compliance report upload for disconnected environments?

26.3 Questions for procurement/legal

- Provide all current Broadcom transaction documents, quotes, order confirmations and reseller pass-through terms.
- Provide legacy VMware ELA, Product Guide references, EULA/Software Exhibit and SnS records if the estate contains historical entitlements.
- Which legal entities are covered by the current agreement?
- Are there special negotiated terms around renewal, support, cloud, portability, price protection or hosting?
- What is the Broadcom Site ID and entitlement owner?
- Are there any open disputes or vendor communications about VMware rights?
- Was any entitlement obtained through OEM, CSP, MaaS, hyperscaler or bundled hardware route?
- Who is authorized to speak to Broadcom/reseller during audit or self-verification?

27. Risk Register Template

Risk ID	Risk statement	Likelihood	Impact	Control / remediation
VMW-R01	ESXi cores undercounted because 16-core-per-processor minimum not applied	High	High	Host-level CPU/core workbook with reviewer sign-off
VMW-R02	vSAN capacity exceeds included VVF/VCF entitlement	Medium	High	vSAN TiB calculation and add-on purchase/remediation
VMW-R03	VCF Operations manages non-VCF licensed cores	Medium	High	Operations monitored-object mapping
VMW-R04	VVF deployed in restricted cloud/hosted environment	Low/Medium	High	Cloud/hosting agreement review
VMW-R05	Version 9 compliance report not submitted on time	Medium	High	180-day reporting calendar and evidence repository
VMW-R06	FNMS vCenter data stale, causing wrong license position	Medium	Medium/High	vCenter Inventory Issues report and rerun beacon task
VMW-R07	Legacy perpetual licenses assumed to provide unsupported version rights	Medium	High	Support/upgrade path validation
VMW-R08	Cloud portability assumed for ineligible OEM/CSP license	Low/Medium	High	License source and certified cloud validation
VMW-R09	Retired hosts still counted in renewal quote	High	Medium/High	Decommission evidence and quote challenge
VMW-R10	No saved copy of applicable SPD at order/install date	Medium	Medium	Document repository with dated PDFs

Risk scoring guide

Score	Likelihood	Impact
Low	Unlikely or limited scope with clear evidence	Low commercial/operational exposure
Medium	Known data gap or plausible issue	Could affect renewal, audit or support
High	Confirmed gap or high uncertainty in major estate area	Likely financial, audit, operational or support impact

28. Glossary for SAM Consultants

Term	Meaning for VMware SAM
Authorized Use Limitation	The quantity or meter the customer is allowed to use, usually from the transaction document.
Core	Physical computational unit inside a processor; current VCF/VVF licensing counts physical cores with minimum rules.
Processor / CPU	Physical chip/socket. The 16-core minimum applies per processor for VCF/VVF.
TiB	Tebibyte, 2^40 bytes; used for vSAN entitlement/capacity.
Transaction Document	Quote/order/SOW or mutually agreed ordering document; highest priority in Broadcom order of precedence.
SPD	Specific Program Documentation; product-specific terms such as metric, included components, cloud/hosting, reporting.
VVF	VMware vSphere Foundation; enterprise workload platform bundle.
VCF	VMware Cloud Foundation; full-stack private cloud platform.
Solution License Key	vSphere key that can serve as VCF/VVF solution key for certain v8/VCF versions.
License File	Version 9 licensing mechanism used through VCF Operations / Business Services Console.
Compliance Report	Version 9 installed-base/license report required by SPD at defined cadence.
DIPA	Deemed Internal Purpose Application; relevant to hosting restrictions.
Certified Cloud Services	Cloud services approved for VCF portability under current Broadcom rules.
FNMS	FlexNet Manager Suite; SAM tool used for discovery, inventory, contracts, licenses and compliance evidence.
ELP	Effective License Position; comparison of entitlement and consumption with assumptions and remediation.
vSAN raw capacity	Licensing-relevant physical storage capacity; not necessarily the same as usable/free capacity shown to admins.

29. Sample Client Emails and Evidence Requests

29.1 Evidence request email

```
Subject: VMware Licensing Review - Evidence Request

Hi Team,

As part of the VMware licensing review, please provide the following evidence for the agreed scope:

1. Current Broadcom transaction documents, quotes, order confirmations and support dates.
2. Broadcom portal entitlement export, license keys and any VCF/VVF 9 license files.
3. Applicable VCF/VVF Specific Program Documentation used for the current order or installed version.
4. vCenter list, ESXi host export with sockets/cores, cluster mapping and lifecycle status.
5. vSAN capacity evidence by cluster, including raw/licensing-relevant capacity.
6. NSX, VCF Operations, Automation, Operations for Networks, HCX and VKS deployment details.
7. FNMS VMware inventory exports and vCenter Inventory Issues Analysis report, if available.
8. Any cloud, hosting, DR, migration or service-provider use cases.

Please provide the data as of [DATE] and identify the owner who can sign off each data set.

Regards,
[SAM Consultant]
```

29.2 Stakeholder sign-off wording

```
I confirm that the attached vCenter/ESXi inventory represents the active VMware host estate for the
agreed scope as of [DATE]. I also confirm that the socket/core data and cluster mapping are accurate
to the best of my knowledge. Exceptions, migration hosts and retired hosts are listed in the exception
log.
```

29.3 Data limitation wording

```
The VMware ELP uses FNMS as the primary SAM evidence platform; however, for Broadcom-era VCF/VVF
subscriptions, the final compliance calculation applies the product-specific SPD rules outside FNMS
where required. Manual adjustments are documented in the adjustment log, with source evidence and
owner sign-off.
```

30. Final Senior Consultant Summary

A shallow VMware licensing guide stops at "count cores." A senior SAM consultant goes further: contract hierarchy, SPD version control, core minimums, vSAN TiB, component scope, cloud/hosting restrictions, version 9 reporting, FNMS data quality, audit evidence and renewal optimization.

Senior principle	Practical behavior
Entitlement is legal, not only portal keys	Use transaction documents, agreements and SPDs as source of truth
Consumption is physical infrastructure, not VM count	Calculate ESXi host cores per CPU with 16-core minimum
vSAN is its own risk	Calculate TiB separately and compare to included/add-on entitlement
Bundles have boundaries	Do not use VCF/VVF components outside licensed core scope
Tools need interpretation	Use FNMS as evidence, then apply current Broadcom terms
Version matters	v8 keys, v9 license files and reporting obligations differ
Audit defense requires evidence	Store dated exports, source documents, formulas and sign-offs
Renewal is architecture + licensing	Optimize hosts, clusters, vSAN and bundle selection before quote acceptance

Source References

The guide uses official Broadcom/VMware and Flexera sources where possible. Always validate the latest applicable documents for a real customer, especially because VMware/Broadcom terms can change by date, version and transaction document.

ID	Source	URL
S1	Broadcom / VMware Cloud Foundation Blog - End of availability of perpetual licensing and SaaS services	https://blogs.vmware.com/cloud-foundation/2024/01/22/vmware-end-of-availability-of-perpetual-licensing-and-saas-services/
S2	VMware Cloud Foundation (VCF) Specific Program Documentation, November 2025	https://ftpdocs.broadcom.com/cadocs/0/contentimages/VCF_SPD_November2025.pdf
S3	VMware vSphere Foundation (VVF) Specific Program Documentation, November 2025	https://ftpdocs.broadcom.com/cadocs/0/contentimages/VVF_SPD_November2025.pdf
S4	Broadcom End User Agreement / Foundation Agreement, Version No. 05112026	https://docs.broadcom.com/doc/end-user-agreement-english
S5	Broadcom KB 313548 - Counting Cores for VCF, VVF and vSAN	https://knowledge.broadcom.com/external/article/313548/counting-cores-for-vmware-cloud-foundati.html
S6	Broadcom KB 312202 - License Calculator for VCF, VVF and vSAN	https://knowledge.broadcom.com/external/article/312202/license-calculator-for-vmware-cloud-foun.html
S7	VMware Cloud Foundation Blog - Licensing in VMware Cloud Foundation 9.0	https://blogs.vmware.com/cloud-foundation/2025/06/24/licensing-in-vmware-cloud-foundation-9-0/
S8	VMware vSphere Product Line Comparison 9.1	https://www.vmware.com/docs/vmw-datasheet-vsphere-product-line-comparison
S9	FlexNet Manager Suite Documentation - VMware Inventory	https://docs.flexera.com/fnms/discovery/dev-inventory-landing/v-mware-inventory
S10	FlexNet Manager Suite Documentation - VMware Device and Infrastructure Inventory	https://docs.flexera.com/fnms/discovery/vir-virtual-devices-landing/vir-virtual-devicesand-clusters/dev-v-mware-device-infra
S11	FlexNet Manager Suite Documentation - vCenter Inventory Issues Analysis Report	https://docs.flexera.com/fnms/reports-introduction/reports-ootb/v-center-inventory-issues-analysis-report
S12	Flexera FlexNet Manager Suite for Enterprises datasheet	https://www.flexera.com/sites/default/files/datasheet-fms.pdf
S13	Broadcom KB 319282 - VCF and VVF Solution License Keys	https://knowledge.broadcom.com/external/article/319282/vmware-cloud-foundation-and-vsphere-foun.html
S14	Broadcom KB 281797 - Upgrade and Downgrade VMware License Keys	https://knowledge.broadcom.com/external/article/281797/upgrade-and-downgrade-vmware-license-key.html
S15	Broadcom KB 432439 - vSphere 8 Licensing Error: License capacity is insufficient	https://knowledge.broadcom.com/external/article/432439/vsphere-8-licensing-error-the-license-c.html